ACE Cyber Preliminary Literature Map: Fall 2025

1. 1. Project Information

Student Name(s): _Denzel Jefferson_____

Project Title: __Contextualized Threat Intelligence Prioritization for CTI Analysts_____

Date / Version: ___May 4, 2026_____

Mentor/Advisor: ____Professor William Littleton_____

2. Research Focus

a. Research Question: How can operational environment data gathered from Elasticsearch be used to automate and improve threat intelligence prioritization within CTI workflows?

b. Generate a list of detailed “Keywords” and “Key Concepts”:
(Example: phishing, human factors, awareness training, cyber resilience, social engineering)

- Threat intelligence prioritization
- CTI analysts
- Alert fatigue
- CTI analyst efficiency
- Automated threat prioritization
- Elasticsearch operational data
- MISP-based threat intelligence scoring (simulated)
- Security Information and Event Management (SIEM)
- Integration of operational data with threat intelligence
- Threat importance ranking
- Event correlation (patterns and root causes)
- CTI Prioritization Workflow
- Automated IOC prioritization
- Context enrichment
- Indicator of Compromise (IOC) scoring
- Threat intelligence feeds
- Security Orchestration, Automation, and Response (SOAR)
- Weighted contextual scoring model

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3. Supporting Literature Table

Citation (APA/IEEE)	Research Type (Empirical, Review, etc.)	Key Findings	Relevance to Research Question	Gap/Limitations Identified
Author, Year, Title	Empirical Study	Summary in 2–3 sentences	Why it matters to your project	What's missing / unresolved
Campbell, C. (2025, April 29). 5 Ways Cyber Threat Intelligence Boosts Cyber Risk Prioritization. Bitsight. https://www.bitsight.com/blog/5-ways-to-boost-cyber-risk-prioritization-with-cti		States that contextual threat intelligence can enhance prioritization by combining business risk and asset detection. When enhanced, CTI will reduce risks and support better detection. Acknowledges the automated system, but does not include applied experimental validation	Supports the idea that contextual information improves prioritizatio n systems.	Not Empirical, doesn't talk about using automated systems or related to tools, including MISP, and Elasticsearch.
Miller, J. (2025, November 4). Automated threat neutralization: How it works and why it matters. BitLyft Cybersecurity. https://www.bitlyft.com/resources/automated-threat-neutralization-how-it-works-and-why-it-matters		Talks about how automated systems detect, identify, and respond to threats faster than a manual process.	Relates to the project's automated approach and states that it can reduce the manual process to improve efficiency.	Information comes from a conceptual discussion without applied validation

ACE Cyber Preliminary Literature Map: Fall 2025

Team, K. C., & Kapon, B. (2024, December 12). The role of a threat intelligence analyst: Kela Cyber. Kela Cyber. https://www.kelacyber.com/blog/the-role-of-a-threat-intelligence-analyst/		Explains CTI analyst responsibilities, including the decision-making process. Also describes how CTI analysts handle large amounts of information and the need to filter out distracting noise.	Provides good detail on the problems CTI analysts face and potential solutions.	Not research-related, and it does not evaluate technical solutions or implementations
Miranda, P. (2025, July 30). What is Security Information and Event Management (SIEM). CrowdStrike. https://www.crowdstrike.com/en-us/cybersecurity-101/next-gen-siem/security-information-and-event-management-siem/		Discusses the functions and details of SIEM, including data collection, alerts, and data analysis. States that SIEM will help CTI analysts prioritize essential alerts.	Provides a description of SIEM data and how data from SIEM tools, such as Elasticsearch, is used in threat prioritization.	Very descriptive but lacks in the evaluation, or is not directly specific to automated prioritization systems.
Baker, K. (2025, March 4). Cyber Threat Intelligence Explained. CrowdStrike. https://www.crowdstrike.com/en-us/cybersecurity-101/threat-intelligence/		Provides details on Cyber Threat Intelligence (CTI), including its importance, benefits, and purpose. Seeks to improve threat awareness and transform raw data into actionable information.	Gives insight into what CTI is and why it's important to use while discussing prioritization improvements.	A broader review of what CTI is, with no mention of automated tools or any workflows.

ACE Cyber Preliminary Literature Map: Fall 2025

Microsoft. (n.d.). *What is Siem?: Microsoft security*. What Is SIEM? | Microsoft Security. https://www.microsoft.com/en-us/security/business/security-101/what-is-siem		Provides a definitive definition of SIEM and its methods for logging event data. While also discussing the role of SIEMs in the cybersecurity field with real-time visibility.	Shows a clear understanding of what operational environment data is and details contextual prioritization.	Mostly general information with no prioritization research.

5. Preliminary Themes & Gaps

Emerging Themes:
- _Contextual data improves threat prioritization accuracy_____
- _Automation is crucial to reduce CTI analysis workload and alert fatigue_____
- _CTI analysts consistently struggle with high data volumes and disorganized prioritization_____

Potential Research Gap:
___While existing research discusses threat intelligence prioritization, automation, and SIEM context, there is a lack of applied research demonstrating how operational environment data from Elasticsearch can be integrated into a contextualized threat prioritization model. In addition, there is limited experimental validation showing how such systems improve CTI analyst efficiency and reduce alert fatigue._____

6. Next Steps

What additional areas will you search/explore in the next 2 weeks?

- Log generation and techniques for Elasticsearch

- Potential case studies regarding the combination of SIEM and threat intelligence platforms
- Existing threat intelligence prioritization workflows and automation techniques
- Research on alert fatigue and CTI analyst performance stats
- Academic studies on MISP integration with SIEM platforms, automated IOC scoring algorithms, and contextualization of threat data deriving from log sources

What gaps need more sources?

- Tool-specific research involving rule-based or automated prioritization workflows
- Evaluations of comparison between automated threat scoring systems
- Empirical studies assessing CTI analysis efficiency improvements

Any difficulties so far?

- Limited academic research on the integration of operational log data with threat prioritization systems
- Most sources are descriptive in nature and not empirical (based on experiments and observations)
- Difficulty finding studies involving realistic log generation for testing